



GE VERNOVA

**ADMS**

16.25.1

Permission Management Plug-In User Guide

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## Change History

| Date         | Change Description                                                                                                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan 26, 2023 | ADMS 3.16: <ul style="list-style-type: none"><li>• Editorial and formatting updates.</li></ul>                                                                                    |
| Oct 20, 2023 | ADMS 16.1.0: <ul style="list-style-type: none"><li>• The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.</li></ul> |
| Apr 16, 2024 | ADMS 16.7.0: <ul style="list-style-type: none"><li>• Updated document format and styling.</li></ul>                                                                               |

# Contents

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>1. Introduction</b>                                  | <b>4</b>  |
| 1.1. Terminology                                        | 4         |
| <b>2. UI Components</b>                                 | <b>6</b>  |
| 2.1. Tree Nodes                                         | 6         |
| 2.2. Three-State Check Boxes                            | 6         |
| 2.3. Pending Status Indicator                           | 7         |
| 2.4. Permission Controls                                | 7         |
| 2.5. Permission Mode                                    | 8         |
| 2.6. Node Text                                          | 9         |
| 2.7. Legend                                             | 9         |
| 2.8. Default View                                       | 10        |
| 2.9. Message Log                                        | 10        |
| <b>3. Getting Started</b>                               | <b>12</b> |
| 3.1. Loading the Plug-In                                | 12        |
| 3.2. Modifying Permissions                              | 12        |
| 3.3. Applying and Canceling Permissions                 | 14        |
| <b>4. Configuring the UI</b>                            | <b>16</b> |
| 4.1. Modifying Settings                                 | 16        |
| 4.2. Disabling Audible Alarms                           | 16        |
| 4.3. Customizing Colors                                 | 17        |
| 4.4. Customizing Fonts                                  | 18        |
| 4.5. Other Settings                                     | 19        |
| 4.6. Configuration File                                 | 20        |
| 4.6.1. Saving the Configuration File                    | 20        |
| 4.6.2. Sample Configuration File                        | 21        |
| 4.6.3. Functional AOR for Saving the Configuration File | 22        |
| 4.7. Restoring Default Values                           | 22        |
| <b>5. Detached Areas</b>                                | <b>23</b> |
| 5.1. Purpose of Detached Areas                          | 23        |
| 5.2. Criteria for Detaching an Area                     | 23        |
| 5.3. Propagating Permissions                            | 23        |
| <b>6. Functional AORs</b>                               | <b>25</b> |
| 6.1. Description of Functional AORs                     | 25        |
| 6.2. List of Functional AORs                            | 25        |
| 6.3. Hierarchical Functional AORs                       | 25        |
| 6.3.1. Description of Hierarchical Functional AORs      | 26        |
| 6.3.2. Inheriting Hierarchical Functional AORs          | 26        |

# 1. Introduction

This chapter introduces concepts used later in the document.

## 1.1. Terminology

The following terms and acronyms are used throughout this document:

- **Area of Responsibility (AOR):** An Area of Responsibility represents an entity that can be assigned to a user. For an electrical system, a device is typically assigned to an AOR, and the AOR is assigned to a user.

An AOR can also represent functionality that a user may or may not be allowed to perform. In this case, a software function is assigned to an AOR, and the AOR is in turn assigned to a user.

The concept of an AOR generally matches the current usage of an AREA record in the Habitat PERMIT database. These terms are used interchangeably within this document.

- **Area of Responsibility Reference (AORRef):** The user interface (UI) displays user permissions in a hierarchical, tree view format. Each node in the hierarchy is an AORRef. An AORRef can serve either of two purposes, which are not mutually exclusive:
  - An AORRef can point to or "reference" an AOR. This is the primary means for binding how the AORs are displayed on the UI to the underlying AREA records in PERMIT.
  - An AORRef can serve as a container for other AORRefs, allowing for the hierarchical organization of AORs in a tree view format.

AORRefs are modeled as HMODE records in the PERMITNV database.

- **Power System AOR:** A power system AOR is an AOR that represents a part of the electrical system. For example, a north AOR and a south AOR apply to electrical equipment in the north and south parts of the system, respectively.
- **Functional AOR:** A functional AOR is an AOR that relates to functional capability, as opposed to being tied to specific equipment. For example, the AUDTEMP functional AOR controls the user's ability to temporarily disable audible alarms.
- **Detached AOR:** A detached AOR is a power system AOR that is no longer available for routine assignment to users. Detached AORs represent portions of the distribution network that have been assigned to specific entities (such as field personnel), usually during storm restoration efforts.
- **Internal AOR:** As described above, AORRefs can be containers for other AORRefs, as well as pointers to AREA records in PERMIT. If an AORRef plays both roles, the AORRef is modeled as two distinct HMODE records in PERMIT. The first HMODE represents the container. The second HMODE exists as a child of the container; it points to an AREA record in PERMIT, and this AREA record is termed an "internal AOR".

By modeling internal AORs in this way, all AREA records exist as end-nodes or "leaves" of the

tree. From a server perspective, this has many advantages. However, the UI can be configured to show or hide internal AORs.

## 2. UI Components

This chapter describes the various components of the Permission Management plug-in.

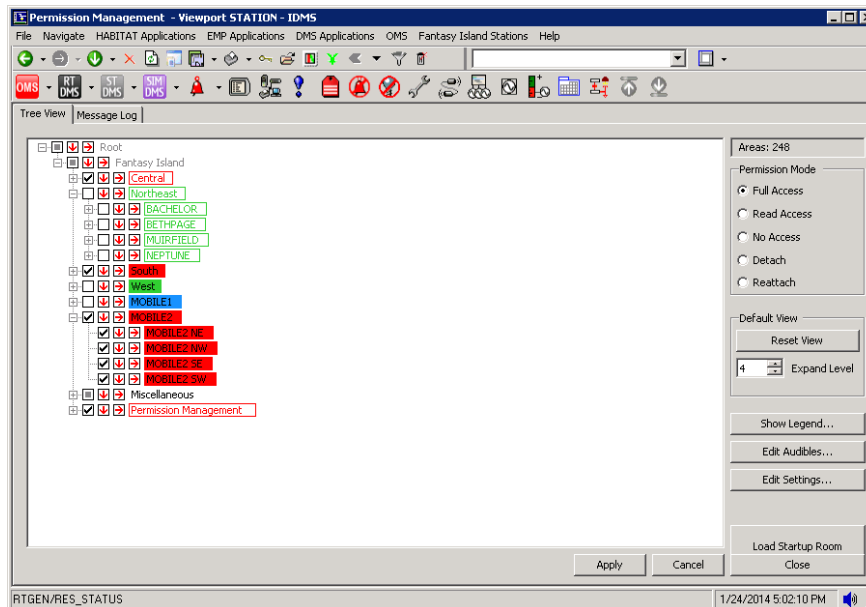


Figure 1. Permission Management Window

## 2.1. Tree Nodes

The main window consists of a collection of nodes, arranged in a tree-like structure. Each row in the tree corresponds to a single AORRef.

In the following image, the AORRef labeled "Baker" is circled in red.

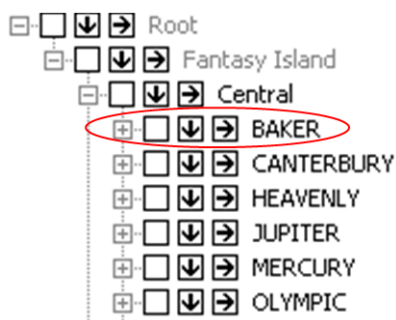


Figure 2. "Baker" AORRef Label

## 2.2. Three-State Check Boxes

Each tree node consists of a series of boxes followed by a text label. The first of these boxes is a three-state check box.

The three possible check box states are: checked, unchecked, and grayed. The check box state indicates the level of permissions assigned to the given area and its descendants. These states are summarized in [Table](#).

#### Note

The three-state check boxes are only visual indicators. Selecting a check box has no effect on the display.

*Table 1. Check Box States*

| Image                                                                             | State     | Description                                                                                                                             |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
|  | Checked   | Indicates that the area and all descendants have full access                                                                            |
|  | Unchecked | Indicates that neither the area nor any of its descendants have full access                                                             |
|  | Grayed    | Indicates that the area is neither checked nor unchecked (i.e., one or more areas have full access, but not all areas have full access) |

## 2.3. Pending Status Indicator

If the permissions for a given AORRef have been changed by the user, but the permissions have not yet been applied by the server, the background color of the check box changes to indicate that the permissions are in the "pending" state. By default, the color for the pending status is yellow.

After the permissions have been applied by the server, the server sends a message back to the client. When the plug-in receives this callback message, the pending status indicator is removed from the UI.

#### Note

The pending status color of a check box is configurable. The colors in [Table](#) are based on the default values.

*Table 2. "Pending" Check Box States*

| Image                                                                               | State             | Description                                                                  |
|-------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------|
|  | Pending Checked   | Indicates that the area is in the checked state, but the status is pending   |
|  | Pending Unchecked | Indicates that the area is in the unchecked state, but the status is pending |
|  | Pending Grayed    | Indicates that the area is in the grayed state, but the status is pending    |

## 2.4. Permission Controls

Each tree node contains one or more permission controls, which appear to the right of a check box. These controls are the primary mechanism for changing a user's permissions. For example, to request full access for a given AORRef and all of its descendants, select Full Access from the

Permission Mode list (refer to [Permission Mode](#)), and click on the Full Down image (red down arrow). [Table](#) summarizes the behavior of these controls.

### **Note**

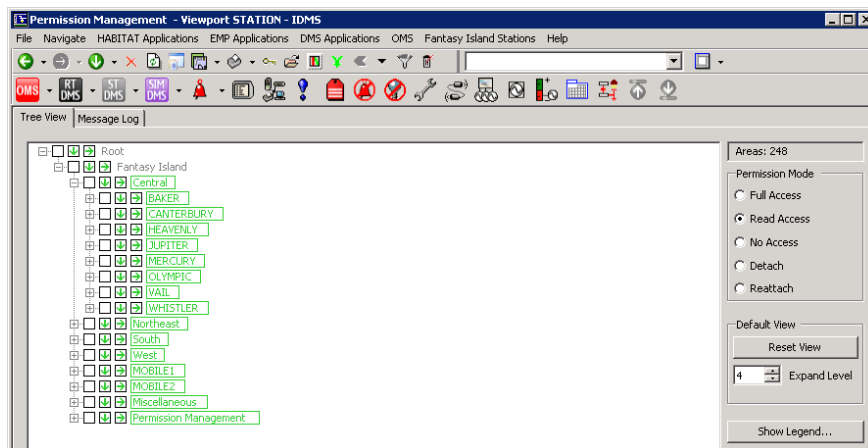
These colors are configurable. The colors in the following table are based on the default values.

*Table 3. Permission Controls*

| Image                                                                             | State      | Description                                                    |
|-----------------------------------------------------------------------------------|------------|----------------------------------------------------------------|
|  | Full Down  | Requests full access for the selected area and all descendants |
|  | Full Right | Requests full access for only the selected area                |
|  | Read Down  | Requests read access for the selected area and all descendants |
|  | Read Right | Requests read access for only the selected area                |
|  | None Down  | Requests no access for the selected area and all descendants   |
|  | None Right | Requests no access for only the selected area                  |
|  | Detach     | Detaches the selected area and all descendants                 |
|  | Reattach   | Reattaches the selected area and all descendants               |

## 2.5. Permission Mode

In the upper right corner of the Permission Management plug-in, there are radio buttons for selecting the Permission mode. The Permission mode determines which permission controls are visible and configurable on the UI. For example, if the user selects the Read Access option, the Read Down and Read Right permission controls appear on the UI.



*Figure 3. Permission Mode List*

## 2.6. Node Text

The color of the node text and the background color of the node text indicate the user permissions for a given area. These color indicators are described in the following table.

*Table 4. Node Text Colors*

| Image                                                                             | State                | Description                                                                                              |
|-----------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------|
|  | Full                 | Indicates that the user has full access permission for the area                                          |
|  | Full + Detached      | Indicates that the user has full access permission for the area, but the area is detached                |
|  | Read                 | Indicates that the user has read access permission for the area                                          |
|  | Read + Detached      | Indicates that the user has read access permission for the area, but the area is detached                |
| Black Text                                                                        | None                 | Indicates that the user does not have permission for the area                                            |
|  | None + Detached      | Indicates that the user does not have permission for the area, but the area is detached                  |
|  | Read Only            | Indicates that the user has read access to a read-only area.                                             |
|  | Read Only + Detached | Indicates that the user has read access to a read-only area, but the area is detached                    |
| Gray Text                                                                         | No Access            | Indicates that the Access mode for the area is No Access (i.e., the user never gains access to the area) |

## 2.7. Legend

The plug-in includes a legend for displaying the current color scheme of the UI. The legend can be accessed via the Show Legend button on the Tree View tab of the plug-in.

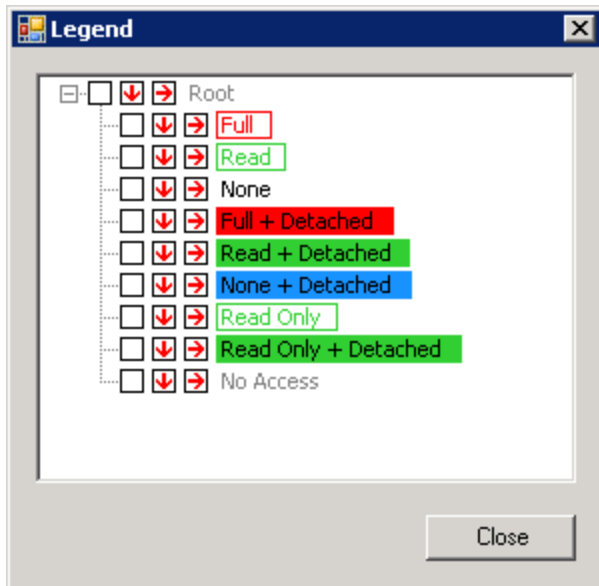


Figure 4. Legend Window

## 2.8. Default View

When a user loads the plug-in, the user is presented with a default view of the hierarchy. The default view is determined programmatically based on the value of the Expand Level parameter. Setting this parameter expands all nodes down to a certain level of the hierarchy. For example, the default Expand Level of 4 expands all nodes in the first four hierarchical levels. All other nodes are collapsed. Clicking the Reset View button resets the Expand Level to the default value.

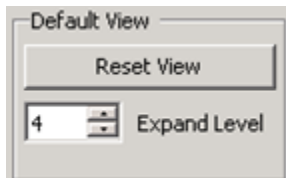


Figure 5. Default View Controls

## 2.9. Message Log

For logging and debugging purposes, the UI includes a Message Log tab for recording the RUSER messages, both pending and sent. At the bottom of the tab is the Clear button, which removes the sent RUSER messages from the UI. For clarity, sent messages are colored black, and pending messages are colored red.

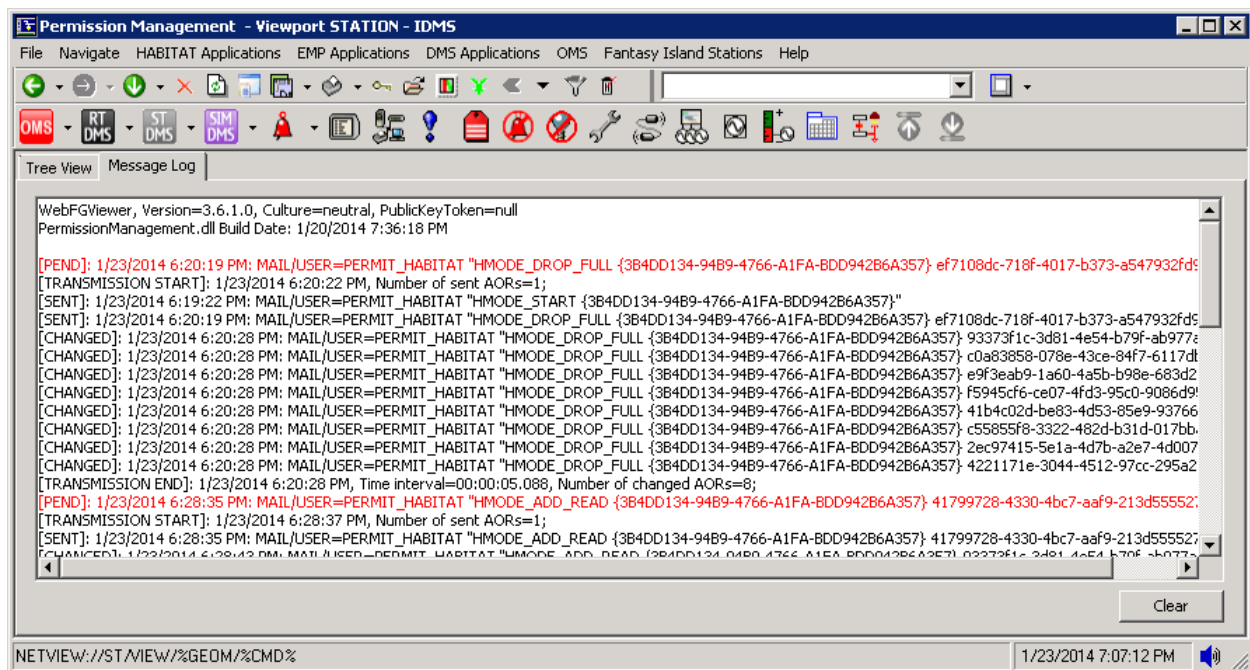


Figure 6. Message Log Tab

## 3. Getting Started

This chapter provides a walk-through demonstration of using the Permission Management plug-in.

### 3.1. Loading the Plug-In

To load the Permission Management plug-in, click on the RT DMS split button , and select Go to Permission Management Plug-In.

Alternatively, the Permission Management plug-in can be added to a startup room.

After the plug-in loads the model from PERMIT, the UI displays the AORRef hierarchy in the main window.

#### Note

The HACTIVE\_ITEMS property must be set to True in the PERMITNV database using HDBRIO for the hierarchical AOR model to appear in the Permission Management plug-in.

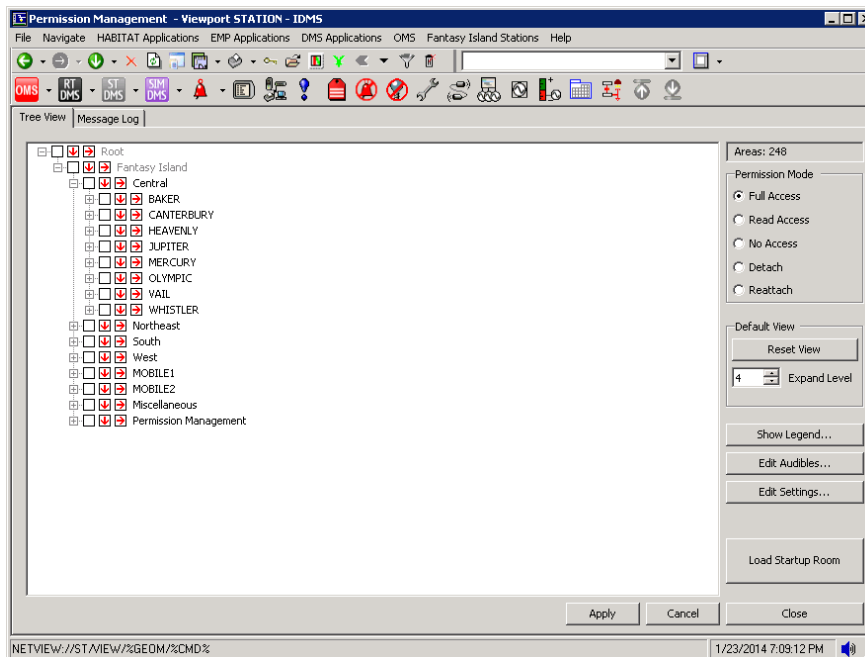


Figure 7. Permission Management Display Showing AORRef Hierarchy

The Areas text box displays the total number of AORs included in the system.



### 3.2. Modifying Permissions

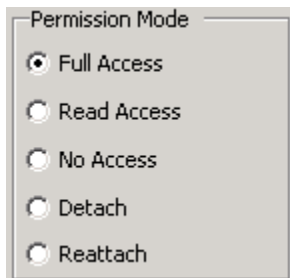


## Important

In the default Fantasy Island permit model, the Hierarchical Mode functionality is disabled. To modify permissions using the Permission Management plug-in, enable the Hierarchical Mode functionality and restart PERMSRV. For more details, refer to the *ADMS Installation Guide*.

### To modify permissions:

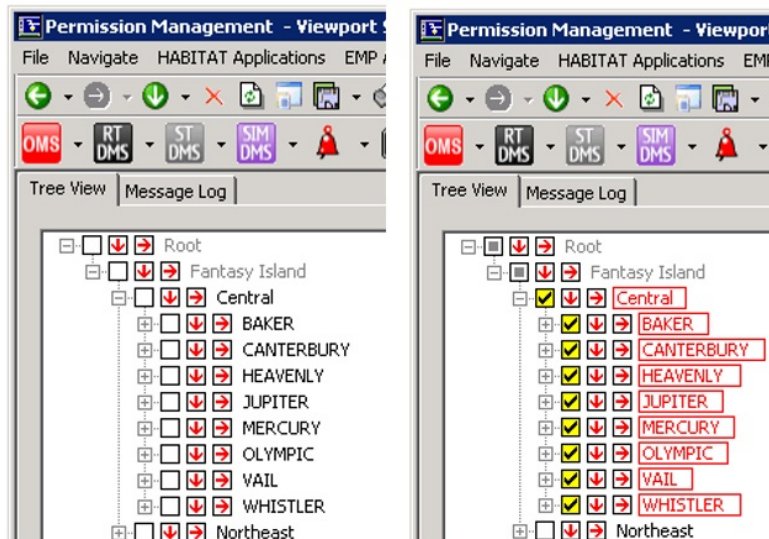
1. In the upper right corner of the viewport, from the group of radio buttons, select the desired Permission mode.



2. In the Tree View list, select one of the permission controls (red arrows).

For example, to request full access to the entire Central branch, click on the red down arrow next to the Central node.

After you select the red down arrow of the Central branch, the UI updates to reflect the pending permission changes. Notice the yellow background of the check boxes for the Central (and children), indicating the pending status.



## 3.3. Applying and Canceling Permissions

To apply the permission changes, click the Apply button in the lower right corner of the viewport.

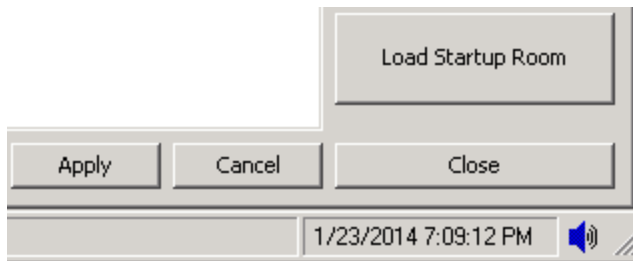


Figure 8. Applying Permission Changes

To cancel the permission changes, click the Cancel button.

After the changes have been applied, notice that the pending status indicators have been removed from the UI.

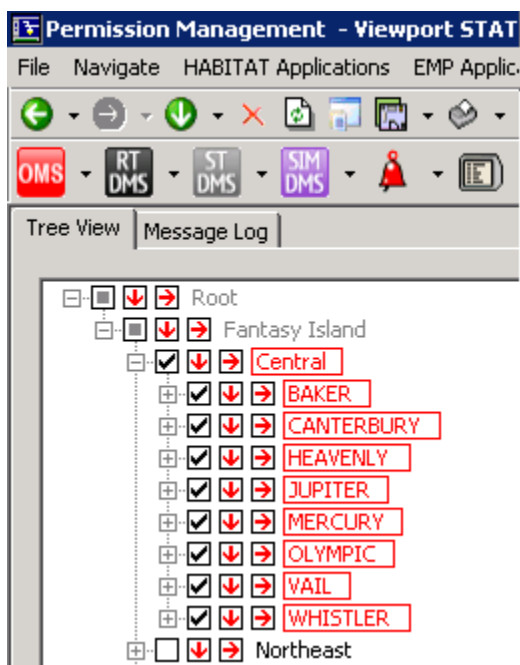


Figure 9. Applied Changes No Longer Show Pending Status

## 4. Configuring the UI

This chapter describes the configurable settings of the Permission Management plug-in.

### 4.1. Modifying Settings

The Permission Management plug-in provides many settings for customizing the UI.

To access these settings, on the Tree View tab of the plug-in, click the Edit Settings button. The Edit Settings pop-up appears.

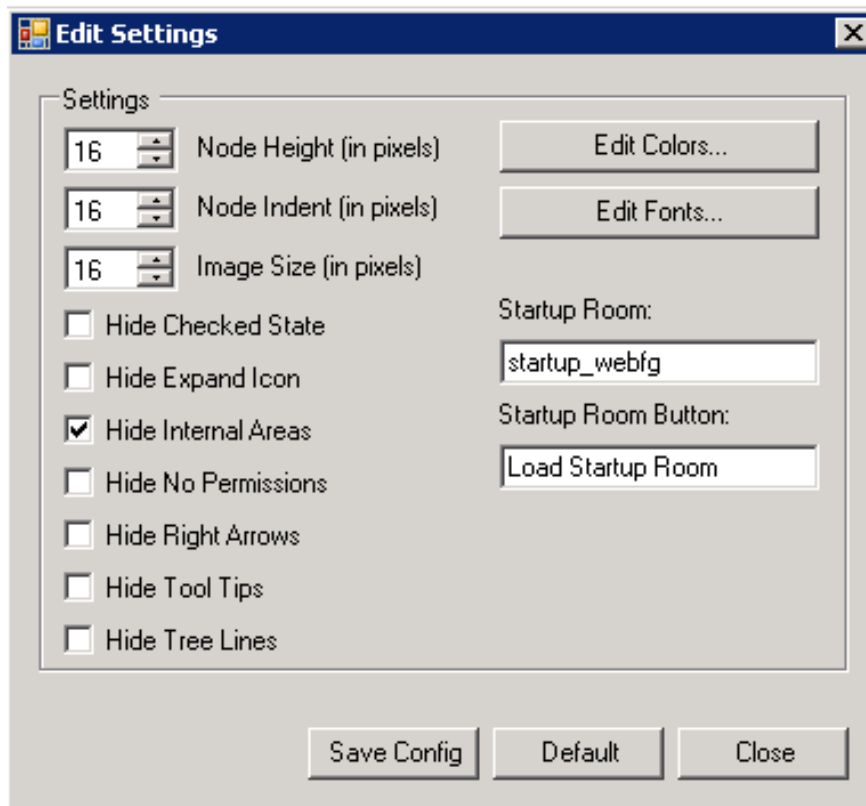


Figure 10. Edit Settings Pop-Up

### 4.2. Disabling Audible Alarms

The plug-in allows the user to disable or re-enable audible alarms, based on the user's permissions for the functional AORs: AUDPERM and AUDTEMP.

To access these settings, click the Edit Audibles button on the Tree View tab of the plug-in.

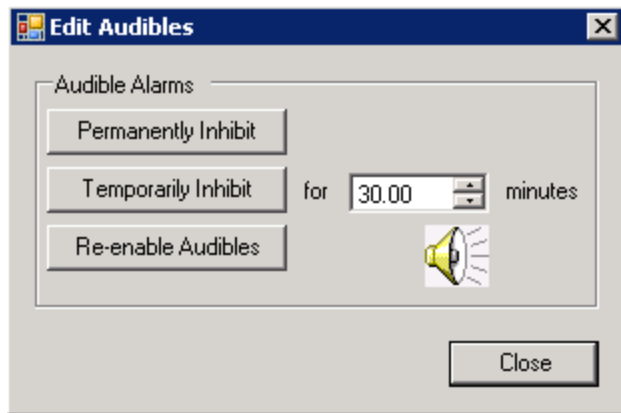


Figure 11. Edit Audibles Pop-Up

If the user has Write permission for the AUDPERM functional AOR, the user can disable audible alarms permanently.

Additionally, the AUDPERM functional AOR determines the default mode for audible alarms. If the Audible flag is TRUE, audibles are enabled at login time. If the Audible flag is FALSE, audibles are disabled at login time.

If the user has Write permission for the AUDTEMP functional AOR, the user can disable audible alarms temporarily for a configurable number of minutes.

For more information about the modeling and usage of functional areas, refer to [Functional AORs](#).

## 4.3. Customizing Colors

The Permission Management plug-in provides a form for customizing the colors of the UI, including:

- Permissions (full, read, none, detach, reattach)
- Node text (read only, no access, text color, text background)
- Check boxes (checked, unchecked, grayed, pending)
- Tree (expand sign, expand background, expand box, tree lines, tree background)

To open the form for customizing colors, click the Edit Colors button on the Edit Settings pop-up.

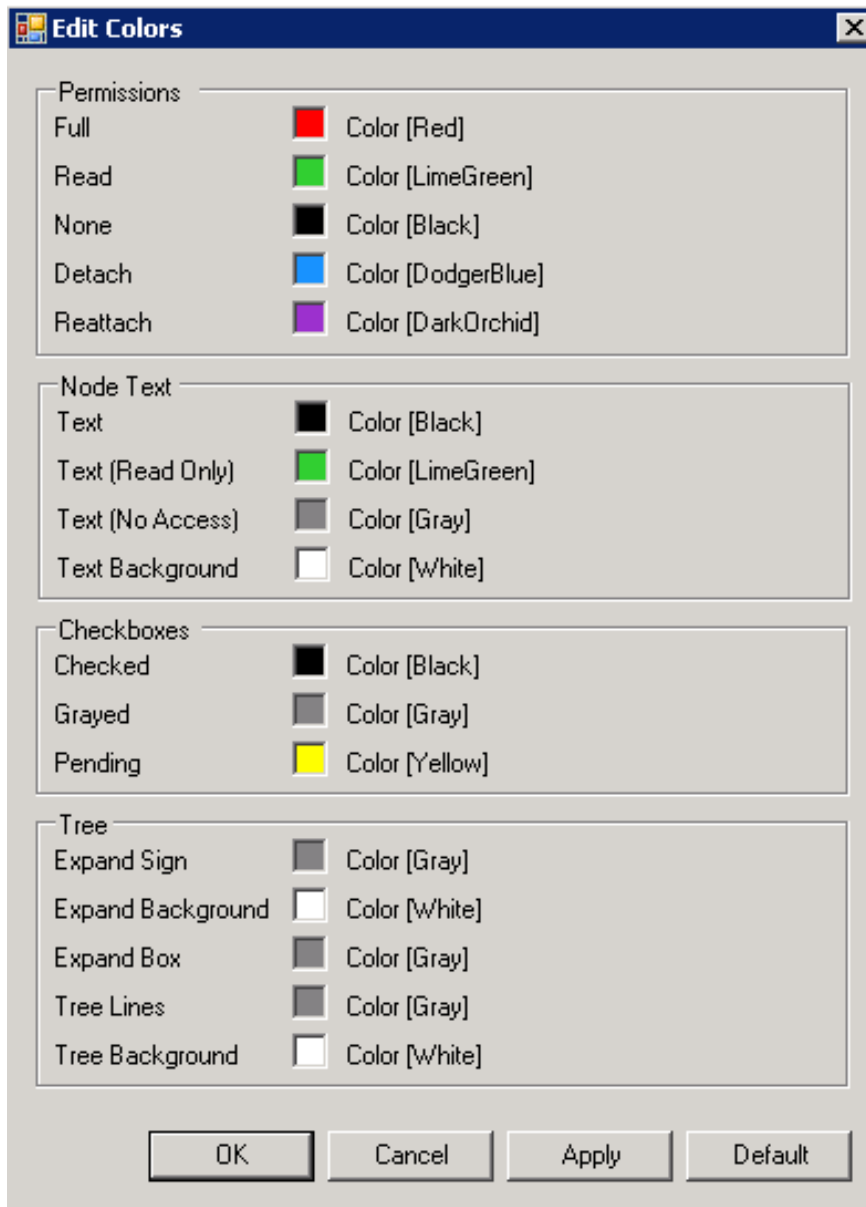


Figure 12. Edit Colors Form

## 4.4. Customizing Fonts

The plug-in provides a standard font dialog box for customizing the font (font type, size, and style) of the AORRef text labels.

To configure fonts, click the Edit Fonts button on the Edit Settings pop-up.

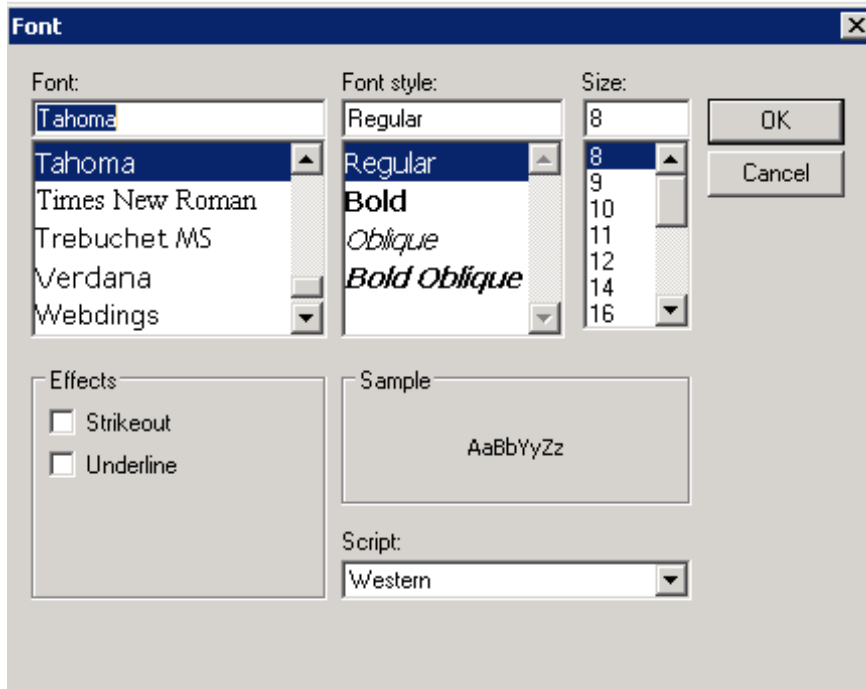


Figure 13. Font Selection Dialog Box

## 4.5. Other Settings

In addition to the parameters described above, the Edit Settings pop-up of the plug-in provides several other customizable parameters, including:

- Node Height (in pixels): The vertical spacing between adjacent rows in the hierarchy
- Node Indent (in pixels): The horizontal spacing between consecutive hierarchical levels
- Image Size (in pixels): The size of the images on the UI (for example, checked state boxes and permission controls)

Since the images are square, the image size corresponds to the length of each side in units of pixels.

- Hide/Show checked state boxes (checked, unchecked, and grayed)
- Hide/Show the Expand icon for expanding/collapsing nodes
- Hide/Show internal areas
- Hide/Show branches of the hierarchy for which the user has no permissions
- Hide/Show right arrows, which are the permission controls for selecting only the area, not the entire branch
- Hide/Show tool tips, which are displayed when the mouse hovers over the node text
- Hide/Show dotted tree lines, which visually connect the parent and child nodes
- Startup Room: The name of the startup room, which can be called up from the Load Startup

Room button on the Tree View tab

## 4.6. Configuration File

The Permission Management plug-in provides the ability to save settings to a configuration file.

### 4.6.1. Saving the Configuration File

To save the configuration file, on the Edit Settings pop-up, click the Save Config button.

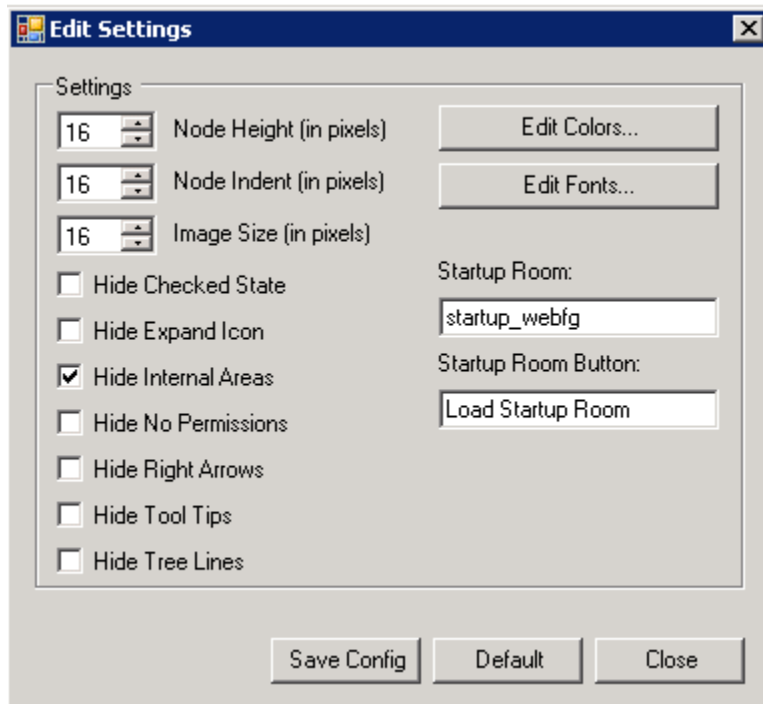
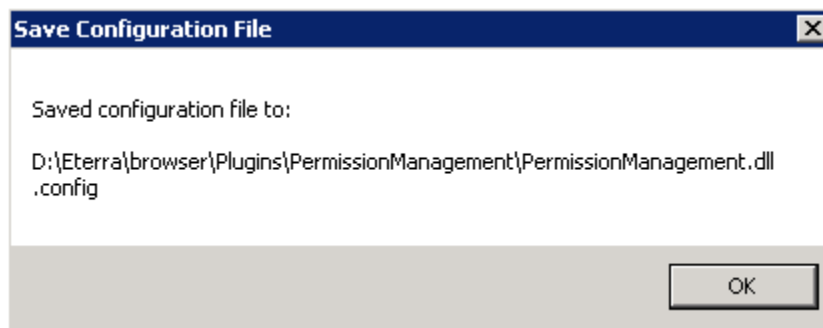


Figure 14. Saving a Configuration File on the Edit Settings Pop-Up

If the save operation is successful, the following message box appears:



The full path name of the configuration file is the same as the PermissionManagement.dll, with the addition of a ".config" extension. For example, if Browser is installed in the D:\Eterra folder, the full

path name of the configuration file is:

```
D:\Eterra\browser\Plugins\PermissionManagement\PermissionManagement.dll.config
```

## 4.6.2. Sample Configuration File

The configuration file is output in XML format. An example of the configuration file appears below:

```
<?xml version="1.0"?>
<PermConfig xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <BoolCheckedState>true</BoolCheckedState>
  <HideExpandIcon>false</HideExpandIcon>
  <HideInternal>true</HideInternal>
  <HideNoPermissions>false</HideNoPermissions>
  <HideRightArrows>false</HideRightArrows>
  <HideToolTips>false</HideToolTips>
  <HideTreeLines>false</HideTreeLines>
  <ExpandLevel>4</ExpandLevel>
  <ImageSize>16</ImageSize>
  <NodeHeight>16</NodeHeight>
  <NodeIndent>16</NodeIndent>
  <StartupRoom>startup_webfg</StartupRoom>
  <NodeFont>Tahoma, 8.25pt</NodeFont>
  <ColorFull>Red</ColorFull>
  <ColorRead>LimeGreen</ColorRead>
  <ColorNone>Black</ColorNone>
  <ColorNoAccess>Gray</ColorNoAccess>
  <ColorReadOnly>LimeGreen</ColorReadOnly>
  <ColorDetach>DodgerBlue</ColorDetach>
  <ColorReattach>DarkOrchid</ColorReattach>
  <ColorChecked>Black</ColorChecked>
  <ColorGrayed>Gray</ColorGrayed>
  <ColorPending>Yellow</ColorPending>
  <ColorExpandSign>Gray</ColorExpandSign>
  <ColorExpandBG>White</ColorExpandBG>
  <ColorExpandBox>Gray</ColorExpandBox>
  <ColorNodeText>Black</ColorNodeText>
  <ColorNodeTextBG>White</ColorNodeTextBG>
  <ColorTreeLines>Gray</ColorTreeLines>
  <ColorTreeBG>White</ColorTreeBG>
</PermConfig>
```

### Note

Although the file can be created or modified manually, the recommended approach for generating a configuration file is through the Permission Management plug-in.

### 4.6.3. Functional AOR for Saving the Configuration File

In order to save a configuration file, the user must have Write permission for the SAVECFG functional area. For more information regarding the use of these special areas, refer to [Functional AORs](#).

## 4.7. Restoring Default Values

To restore the settings to their respective defaults, on the Edit Settings pop-up, click the Default button.

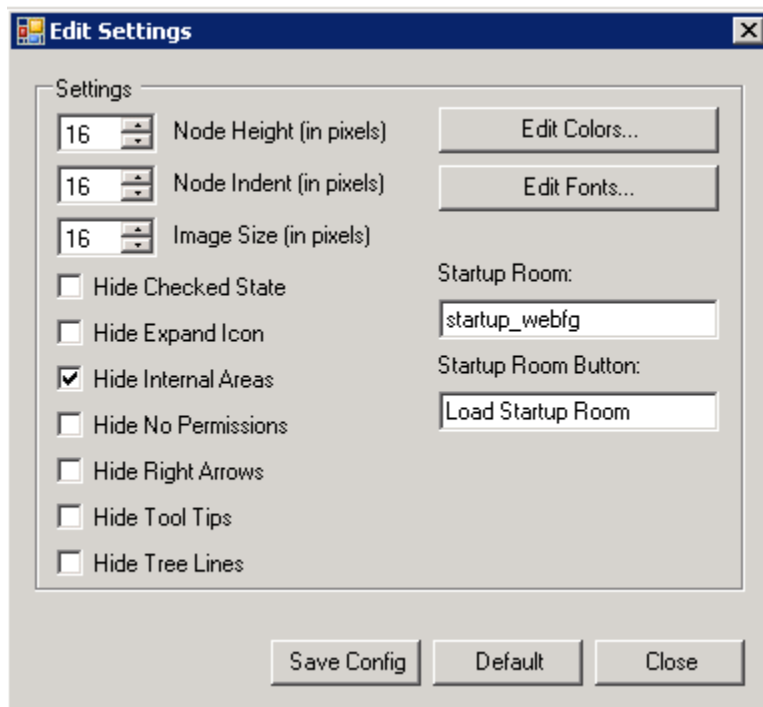


Figure 15. Restoring Default Settings on the Edit Settings Pop-Up

The default values derive from the current configuration file. If the configuration file does not exist, the settings revert to pre-defined values.

## 5. Detached Areas

This chapter elaborates on the concept of detached areas.

### 5.1. Purpose of Detached Areas

Detached areas represent parts of the distribution network that are no longer available for routine assignment to users. For example, an operator might detach certain areas while a crew in the field works on the system.

When a user detaches an area, all other users lose permissions for that area. However, the first user's own permissions for the area do not change.

### 5.2. Criteria for Detaching an Area

The following criteria pertain to detaching areas via the UI:

- In order to detach or reattach an area, the user must have Write permission for the DETACH functional AOR.
- AORRefs are modeled as HMODE records in the PERMITNV database. If the Can Detach field (CANDTCH\_HMODE) is set to FALSE, detaching is denied.
- If the No Access field (NOACCESS\_HMODE) is set to TRUE, detaching is denied.
- If an AORRef is detached, reattaching is denied if the parent is also detached.

### 5.3. Propagating Permissions

Detached areas require special consideration, particularly with regard to the hierarchical assignment of permissions. The following rules summarize these considerations:

- Detaching or reattaching an AORRef always propagates to all descendants.

Note that this does not necessarily mean that all descendants become detached or reattached. Rather, each AORRef is subject to the detaching criteria, as outlined in [Criteria for Detaching an Area](#).

- If a user attempts to detach or reattach an AORRef with the Can Detach field (CANDTCH\_HMODE) set to FALSE, detaching or reattaching does not propagate to the children.

This is consistent with the previous rule, because the selected node must be detachable in order for propagation to start.

- Full access does not propagate from a parent that is not detached to children that are detached.
- Read access and no access propagate to all children, without regard to the detached state of

the children.

## 6. Functional AORs

This chapter describes the use of functional Areas of Responsibility for granting or restricting access to various UI components.

### 6.1. Description of Functional AORs

Fundamentally, there is no difference between a power system AOR and a functional AOR. Both are modeled as AREA records in the PERMIT database. However, functional AORs require special consideration in the operation of the Permission Management plug-in. Some of these considerations are outlined below:

- Generally, functional AORs are restricted to a set of hard-coded, reserved names (the exception being hierarchical functional AORs, which are described later in this chapter).

For example, one of the reserved functional AORs is named "DETACH".

- In most cases, permissions for a functional AOR are controlled by the Write flag. If the Write flag is TRUE, the user has permission to perform the function; otherwise, the user does not have permission to perform the function.
- Functional AORs are modeled in the initial and base modes of the user.
- When a user first loads the plug-in, the plug-in reads the permissions for the functional AORs. These permissions persist for the duration of the session, regardless of whether the permissions change at some point during the session.

### 6.2. List of Functional AORs

*Table 5. Functional AORs and Descriptions*

| Functional AOR | Description                                                                             |
|----------------|-----------------------------------------------------------------------------------------|
| ANNO           | Allows the user to add or remove annotations on the ADMS displays                       |
| AUDPERM        | Allows the user to permanently enable or disable audible alarms                         |
| AUDTEMP        | Allows the user to temporarily enable or disable audible alarms                         |
| DETACH         | Allows the user to detach or reattach hierarchical areas                                |
| READONLY       | Prevents the user from modifying area permissions via the Permission Management plug-in |
| SAVECFG        | Allows the user to save configuration files via the Permission Management plug-in       |

### 6.3. Hierarchical Functional AORs

In addition to the hard-coded functional AORs, each node in the hierarchy (optionally) references a functional AOR.

### 6.3.1. Description of Hierarchical Functional AORs

The purpose of a hierarchical functional AOR is to override the nominal assignment of AORRef permissions. This overriding behavior depends on which of the permissions (read, write, execute, and audible) is set in the functional AOR as contained in the base or initial modes of the user.

The following bullet points describe this behavior explicitly:

- If the Read and Write flags are set to FALSE, the branch of the hierarchy is hidden from the user.
- If the Read flag is set to TRUE but the Write flag is set to FALSE, the branch of the hierarchy is read-only. In other words, none of the AORs contained in the branch can be assigned full access. If the user requests full access at any of these nodes, the plug-in grants only read access.
- If the Write flag is TRUE and the Audible flag is TRUE, the permission changes are processed normally.
- If the Write flag is TRUE and the Audible flag is FALSE, the plug-in sends full access requests as read/write/execute (i.e., full access minus audibles).

### 6.3.2. Inheriting Hierarchical Functional AORs

As described above, hierarchical functional AORs can be blank (null). The following rules summarize the handling of null permissions with regard to these functional AORs:

- If the functional area is null, the AORRef inherits the functional area permissions from the parent.
- The root functional area permissions are defined to be: read, write, and audible are TRUE. Thus, if no hierarchical functional areas are defined, the tree functions normally (i.e., no branches are hidden, and permissions propagate down the tree without modification).
- If the functional area is not null but it points to an AREA that does not exist, the functional area is handled as if the AREA does exist but the user has no permissions (i.e., read, write, and audible are FALSE). Therefore, the AORRef (and all descendants) are hidden from the user.